

Quantitative, statistically robust characterization of solder joint geometry and voids in commercial BGA assemblies

1st Jana Christine Faes

Dept. of Materials, Textiles and Chemical Engineering (MaTCh)
Ghent University
Ghent, Belgium
Jana.Faes@UGent.be

2nd Matthieu Nicolaas Boone

Centre for X-ray Tomography
Ghent University
Ghent, Belgium
Matthieu.Boone@UGent.be

3rd Jaianth Vijayakumar

European Synchrotron Radiation Facility
71 Avenue des Martyrs
38000, Grenoble, France
Jaianth.Vijayakumar@esrf.fr

4th Nicolas Lammens

Siemens Industry Software NV
Leuven, Belgium
Nicolas.Lammens@Siemens.com

5th Wim Van Paepegem

MaTCh, Ghent University
and FlandersMake@UGent-MIRO
Ghent, Belgium
Wim.VanPaepegem@UGent.be

Abstract—Fatigue life predictions for Ball Grid Array (BGA) assemblies depend strongly on local stress and strain fields, yet most models still assume void-free, idealized solder joint geometries. Real assemblies, however, exhibit complex geometric variability and defect clustering, both of which remain insufficiently quantified. This study aims to provide statistically robust, three-dimensional input data needed for physics-based and probabilistic fatigue models.

High-resolution X-ray CT was applied to approximately 1,500 solder joints across two industrial BGA package types, enabling automated 3D geometry extraction, void detection, and spatial statistical analysis. Joint profiles and volumes were reconstructed from mesh-based cross-sections, and void populations were analyzed with respect to count, total volume, and intra-joint position.

The results reveal that all joints contain voids, demonstrating that voiding should be considered an intrinsic feature rather than an exceptional defect. Void distributions are highly non-uniform: over 90% of voids occur within the top or bottom 15% of the joint height, coinciding with pad interfaces where thermo-mechanical stresses are maximal. Although overall geometric variability is small, the larger BGA type shows reproducible location-dependent trends attributed to PCB warpage, and the joint profiles deviate from the idealized barrel shape. A moderate positive correlation between joint volume and void volume further highlights that geometric variability and voiding are statistically coupled and should not be treated as independent inputs.

These findings collectively demonstrate that realistic joint geometries, spatially clustered void distributions, and their statistical correlations must be incorporated into reliability and fatigue models to achieve physically meaningful predictions of solder joint performance in BGA assemblies.

Index Terms—BGA solder joints, X-ray computed tomography, void distribution, geometric variability, fatigue reliability modeling.

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I. INTRODUCTION

Solder joint reliability in BGA assemblies has been widely studied due to its critical role in determining the lifetime of electronic packages subjected to thermal cycling and mechanical loading [1]–[4]. Classical fatigue models for solder interconnects are often based on simplified geometries, often assuming axisymmetric barrel-shaped joints which are free of defects. However, experimental and numerical studies have shown that local stress and strain concentrations are highly sensitive to geometric details and internal defects, particularly near the pad interfaces where crack initiation typically occurs [5], [6].

Voiding in lead-free SnAgCu (SAC) solder joints has been reported to originate from flux outgassing, which is affected by local process variations during stencil printing and reflow [7]. While industry standards such as IPC [8] define acceptance criteria based on maximum void area fractions in two-dimensional cross-sections, the three-dimensional morphology and spatial distribution of voids are rarely quantified at scale. Several studies have demonstrated that large or unfavorably located voids can significantly increase local plastic strain accumulation and reduce fatigue life [9], [10], yet most modeling approaches still represent voiding through a single averaged parameter or neglect it entirely.

The scarcity of large-scale, statistically representative 3D datasets from commercial assemblies has limited the integration of realistic geometric variability into reliability simulations. The present work addresses this gap by combining high-resolution X-ray CT with systematic statistical analysis across approximately 1,500 solder joints from two industrial BGA package types, thereby providing experimentally grounded input for probabilistic and physics-based fatigue models.

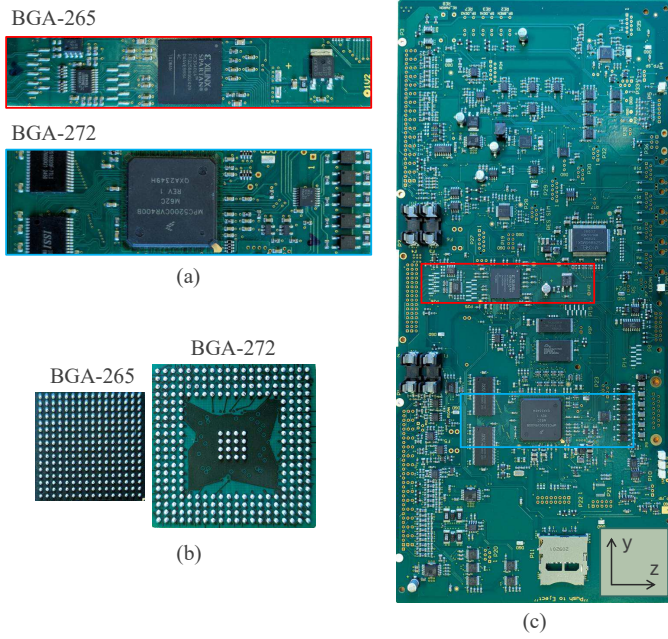


Fig. 1. Investigated PCB sections prepared for computed tomography (CT) scanning (a), corresponding BGA arrays (b), and location of the investigated samples within the full PCB assembly (c).

II. MATERIALS AND METHODS

A. Materials

The samples investigated during this study (Fig. 1.a) each contain a BGA-mounted integrated circuit (Fig. 1.b) and were extracted from a commercially available multilayer printed circuit board (PCB) product (Fig. 1.c). The first device is a Xilinx Spartan field-programmable gate array (FPGA) in a 256-ball BGA package (denoted BGA-256). The second device is a 400 MHz embedded microprocessor in a 272-ball BGA package from NXP Semiconductors (denoted BGA-272).

The BGA-256 package has an approximate body size of $18 \times 18 \text{ mm}^2$ and contains a 16×16 array with nominal solder joint diameters of approximately 0.6 mm and heights of approximately 0.3 mm. The BGA-272 package measures approximately $27 \times 27 \text{ mm}^2$ and contains 272 solder joints with nominal diameters of approximately 0.9 mm and heights of approximately 0.5 mm.

All assemblies were produced using a no-clean lead-free SnAgCu (SAC305) solder paste (Sn–3.0Ag–0.5Cu wt.%). Re-flow soldering was performed using a linear temperature ramp with a total heating duration of approximately 260 s. Peak temperatures reached 242.6°C for BGA-256 and 237.9°C for BGA-272, followed by forced cooling to below 60°C within approximately 100 s.

Three identical PCBs were investigated. From each PCB, one BGA-256 and one BGA-272 section were extracted for CT analysis, resulting in three samples per package type and a total population of approximately 1,500 solder joints.

B. Micro-CT Acquisition and Data Processing

The extracted PCB sections were scanned using the HECTOR micro-CT system [11] at UGCT (Ghent University), configured for applications in electronics inspection. For each package type, the three corresponding samples were mounted in a vertical stack to ensure identical imaging conditions across the set.

The scans were performed at an X-ray tube voltage of 180 kV and a power of 15 W, with a 1 mm copper filter to reduce beam hardening. Each acquisition consisted of 3201 projections with an exposure time of 1000 ms per projection, using detector gain mode 3.

The resulting voxel size was $9 \mu\text{m}$ for the BGA-256 joints and $14 \mu\text{m}$ for the BGA-272 joints, corresponding to approximately 67 and 64 voxels across the joint diameter, respectively. These resolutions are sufficient for reliable characterization of joint geometry and quantification of internal voids.

Volume reconstruction was performed using the standard reconstruction workflow of the HECTOR system. All datasets were reconstructed consistently and exported in 16-bit grayscale format for further analysis.

Subsequent analysis was carried out in Hexagon VGStudioMax [12]. Solder regions and internal voids were segmented using a consistent threshold-based workflow, after which geometric and volumetric parameters were extracted. All quantitative metrics were exported and processed using custom Python scripts. The same processing pipeline was applied to all datasets to guarantee comparability across package types and samples.

C. Optical Microscopy Validation

To assess the reliability of CT-based void detection, selected joints were validated using optical microscopy. For one representative sample of each package type, a row of solder joints was progressively ground and polished to generate sequential cross-sections. Imaging was performed using a Keyence NHX-7000 digital microscope at magnifications up to $\times 500$, enabling clear resolution of void morphology.

The resulting optical sections were aligned with corresponding virtual CT slices using characteristic geometric features. This comparison allowed an assessment of the detection accuracy of voids identified by CT.

D. Synchrotron Micro-CT Validation

To further evaluate the detection limits of the laboratory-based CT system, supplementary high-resolution propagation-based phase-contrast synchrotron micro-CT measurements were performed at the BM18 large-field microtomography beamline [13], [14] at the European Synchrotron Radiation Facility (ESRF).

One sample of each package type (BGA-256 and BGA-272) was stacked and scanned simultaneously. These samples were not part of the statistical dataset analyzed in this study and were therefore used exclusively for qualitative validation.

The measurements were conducted using a polychromatic beam with the peak of the energy spectrum around 140 keV

to reduce streak artefacts in electronic assemblies during reconstruction. Projection images were acquired using a 2× Otus Nikkor optical system coupled to a PCO iB10 detector (4416 × 2360 pixels), yielding an effective pixel size of approximately 2.02 μm. The sample-to-detector distance was set to 1 m.

Reconstruction was performed using in-house TOMWER software [15], applying filtered back-projection combined with Paganin phase retrieval ($\delta/\beta = 100$) to enhance contrast between solder and voids.

The substantially improved spatial resolution compared to the UGCT measurements (9–14 μm pixel size) enables visualization of voids below the laboratory CT detection threshold. These data provide an independent reference for assessing the completeness of void detection and for verifying the general morphology and spatial distribution of the void population observed in the primary dataset.

III. RESULTS

A. Solder Joint Geometry and Volume Distribution

The automatic surface determination tool of Hexagon VGStudioMax was employed to segment the solder material and identify the outer surfaces of each individual joint. The detected surfaces were converted into 3D triangular meshes, and the mesh node coordinates were exported for further processing.

For each joint, horizontal cross sections were extracted along the joint height by projecting the mesh nodes closest to a given plane onto that plane (Fig. 2). The cross-sectional area at each height was calculated using the convex hull of the projected points. Assuming a circular cross section, an equivalent radius was derived from the computed area. The resulting equivalent radius as a function of height represents the average joint profile.

Fig. 3 shows the resulting profile curves for all solder joints in each of the six investigated BGAs. Overall, joints of the same type exhibit limited variability. The BGA-272 packages show a distinct geometric feature in the central 4 × 4 region: these joints are shorter but wider, likely reflecting localized PCB warpage during reflow. This spatial trend is reproducible across all three BGA-272 samples.

Joint volumes were obtained by numerical integration of the profile curves and mapped onto their array positions. The spatial distribution of the calculated volumes is visualized in Fig. 4, where each solder joint is color-coded according to its volume.

Joint volumes were obtained by numerical integration of the profile curves and mapped onto their array positions within the BGA grid. The spatial distribution of the calculated volumes is visualized in Fig. 4, where each solder joint is color-coded according to its volume. Each panel corresponds to a different BGA sample, and the layout of the colored markers reflects the physical grid position of the solder joints within the package.

After exclusion of misidentified solder residues in the BGA-272 samples (cf. Fig. 5), the remaining joint-to-joint volume variations within each package type are small (approximately

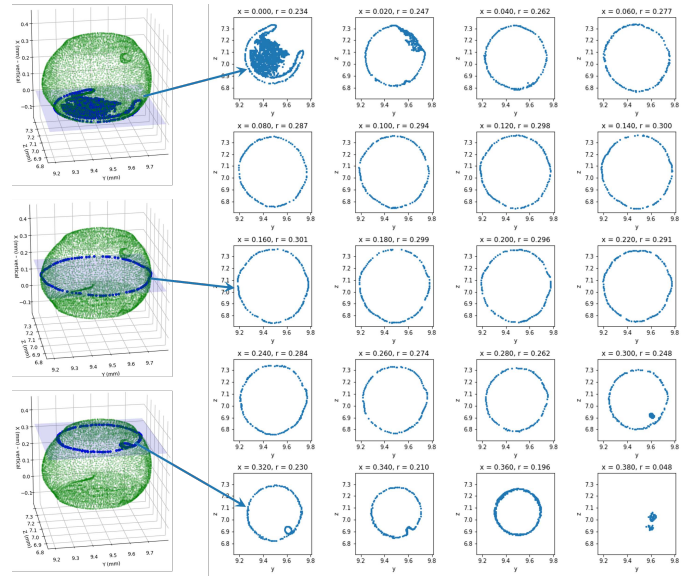


Fig. 2. Horizontal cross sections extracted along the joint height by projecting the mesh nodes closest to a given plane onto that plane.

1%). Given the tight manufacturing tolerances of prefabricated solder balls, these variations are attributed primarily to differences in stencil-printed solder paste volume.

In addition to the small global scatter, a clear location dependency of joint volume is observed for the BGA-272 samples, with systematic differences between central and peripheral array regions. This spatial trend is consistent across all three BGA-272 devices. The clear spatial trend observed in the BGA-272 samples confirms that subtle, process-induced geometric biases are present in these assemblies. Although small in absolute terms, these differences become relevant in the context of void formation, as discussed in the following section.

B. Void Detection Validation

Comparison between CT and optical microscopy revealed partial overlap in detected void populations. For the validated BGA-272 sample in Fig. 6, 19 voids were detected by both methods, 11 only by microscopy, and 5 only by CT. Similar proportions were observed for the BGA-256 sample.

The discrepancies are mainly size-dependent. Very small voids near the CT resolution limit, as well as voids partially merging with the outer joint geometry, were more likely to be missed by the automatic CT-based detection. Conversely, some features identified by CT were not visible in the exact physical cross-section due to slight spatial offsets.

Additional insight into the detection limits of the laboratory CT system is provided by synchrotron measurements. Fig. 7 compares representative cross-sections of a BGA-256 joint reconstructed using the synchrotron setup and the UGCT laboratory system.

The smallest clearly discernible voids in the synchrotron data have diameters of approximately 5 μm, whereas in the

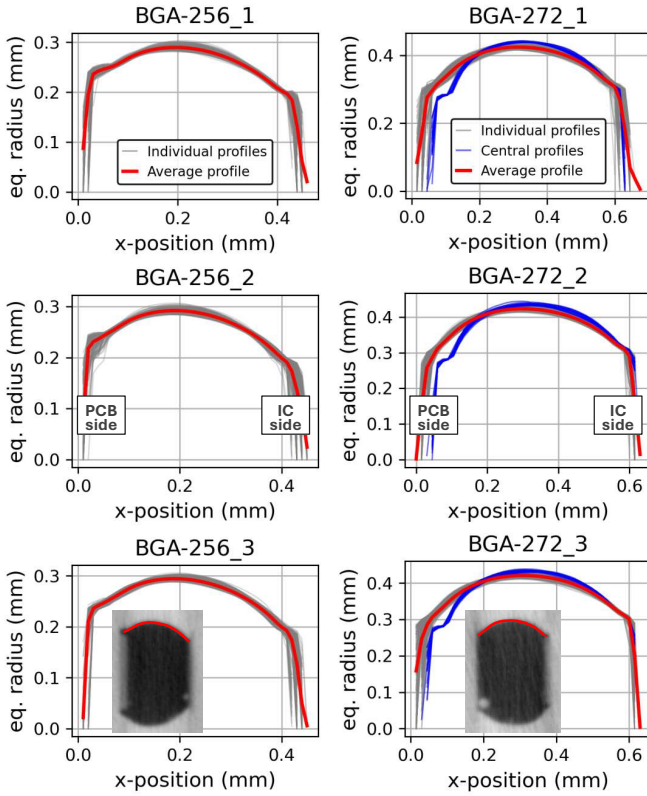


Fig. 3. Joint profile curves for all solder joints in each of the six investigated BGAs, with individual joint profiles in grey, the corresponding average profile in red, and, for the BGA-272 packages, the 4×4 central joints highlighted in blue.

UGCT datasets the smallest reliably detectable voids are about $20 \mu\text{m}$, consistent with the voxel sizes used in both systems.

These results confirm the microscopy observations: voids smaller than the UGCT detection threshold are present in the solder joints, meaning the laboratory CT analysis underestimates the total void count. However, the additional voids visible in the synchrotron data are predominantly very small and contribute only marginally to the total void volume.

Since fatigue crack initiation is primarily driven by larger voids located near high-stress regions, the UGCT data are considered sufficiently accurate for statistical analysis of void volume, spatial trends, and correlations with joint geometry. The synchrotron measurements nevertheless indicate that the absolute number of voids is underestimated due to the limited detectability of the smallest features.

C. Void Count, Volume, and Spatial Trends

Fig. 8 summarizes the number of detected voids per solder joint for all six BGA samples. None of the $\sim 1,500$ solder joints was found to be completely void-free. On average, BGA-256 joints contained approximately 7–8 voids, while BGA-272 joints contained 8–9 voids per joint.

Manual inspection was performed on the most highly voided joints to verify compliance with IPC requirements. According to IPC criteria, the void area must remain below 30% of the

total solder area in any cross-section through the ball [8]. The maximum observed void area fraction was 19% in one BGA-256 sample and 17% in one BGA-272 sample (cf. Fig. 9). All investigated joints therefore remain within IPC acceptance limits.

Fig. 10 depicts the total void volume per joint for all six BGA samples. While void counts do not exhibit a strong location dependency, the total void volume per joint shows a reproducible trend along the z -direction of the array: higher-numbered z -rows consistently exhibit larger total void volumes. No systematic variation is observed along the y -direction, which corresponds to the PCB transport direction during reflow. The origin of this z -direction trend cannot therefore be directly attributed to conveyor-related thermal gradients. It may instead reflect subtle, repeatable asymmetries in local thermal distribution, airflow, stencil printing, or flux outgassing behavior, though further process-specific data would be required to confirm this.

A positive correlation between joint volume and total void volume was expected from physical considerations: larger solder volumes contain more flux and molten material, increasing the probability of gas entrapment before solidification. Correlation analysis, summarized in Table I, confirms this tendency, with statistically significant moderate positive Pearson coefficients for most samples (overall $r \approx 0.3$, $p \ll 10^{-6}$).

D. In-Joint Void Position

The spatial distribution of individual voids within the solder joints was analyzed by plotting each void as a function of its distance to the PCB pad. The results in Fig. 11 show a strong concentration of voids near the pad interfaces, particularly on the component side. More than 90% of all detected voids are located within the upper or lower 15% of the joint height. The central region contains far fewer and generally smaller voids.

This highly non-uniform distribution is mechanically significant, as pad interfaces correspond to regions of elevated thermo-mechanical stress during thermal cycling. Assuming a homogeneous void distribution would therefore underestimate the local stress amplification associated with interfacial void clusters.

IV. CONCLUSION

This study provides a statistically robust three-dimensional characterization of solder joint geometry and voiding in two

TABLE I
PEARSON CORRELATION BETWEEN JOINT VOLUME AND VOID VOLUME.

Sample	r -value	p -value
BGA-256_1	0.0817	1.9×10^{-1}
BGA-256_2	0.2056	9.4×10^{-4}
BGA-256_3	0.4755	7.5×10^{-16}
BGA-272_1	0.4117	1.6×10^{-12}
BGA-272_2	0.3133	1.4×10^{-7}
BGA-272_3	0.2721	6.0×10^{-6}

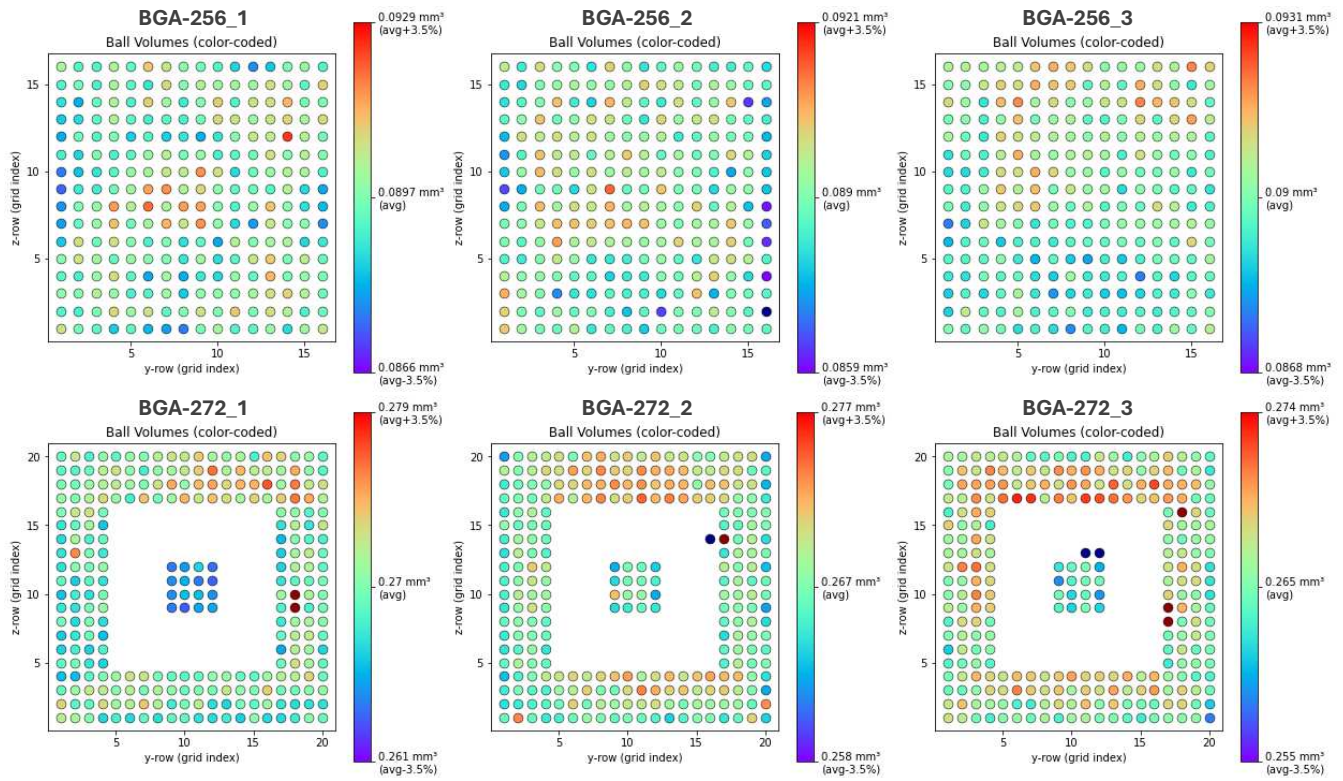


Fig. 4. Spatial distribution of solder joint volumes mapped onto their positions within the BGA grid. Each panel represents a different BGA sample, with each marker corresponding to a single solder joint at its physical array location. The color scale indicates the calculated joint volume.

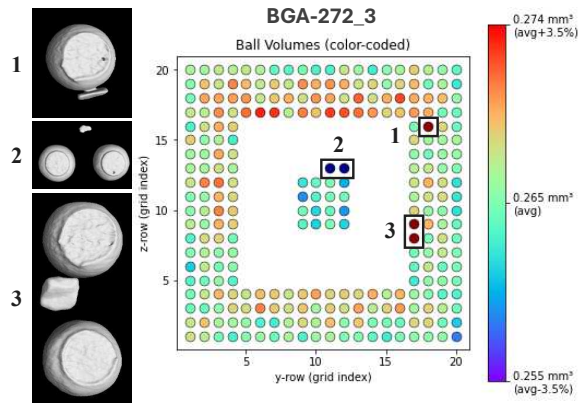


Fig. 5. Localized volume anomalies due to solder residue in sample BGA-272_3 (same sample as shown in Fig. 4). Selected joint locations are highlighted in the array map, and corresponding 3D renderings illustrate the presence of solder residue affecting the reconstructed joint volume.

commercially assembled BGA package types, comprising approximately 1,500 joints. Several findings are directly relevant for reliability modeling and probabilistic fatigue analysis.

First, no solder joint was found to be completely void-free. Voids are therefore an intrinsic feature of the investigated assemblies and should be explicitly accounted for in reliability models rather than treated as anomalies. Models assuming defect-free joints may overestimate fatigue life, especially when crack initiation is governed by local stress concentra-

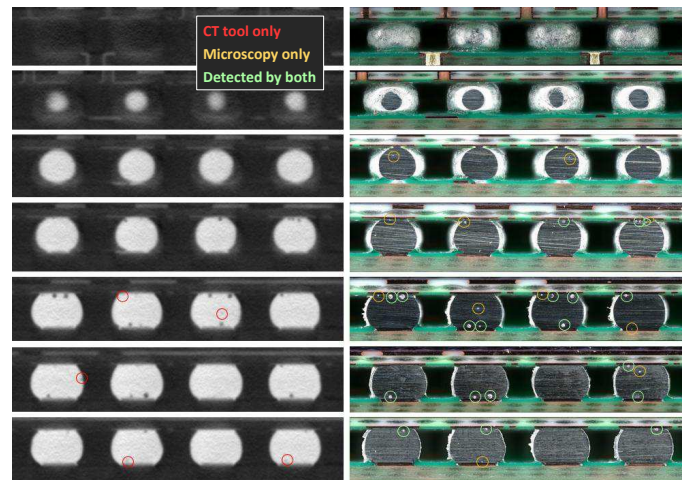


Fig. 6. Void detection comparison for sample BGA-272_1. A cross-sectional CT reconstruction (left) is compared with the corresponding optical microscope image (right), illustrating the overlap in detected void features.

tions.

Second, voids are highly non-uniformly distributed within the joints. More than 90% of all detected voids are located within the upper or lower 15% of the joint height, i.e., close to the pad interfaces. Since these regions experience elevated thermo-mechanical stresses during thermal cycling, assuming a homogeneous void distribution may underestimate local

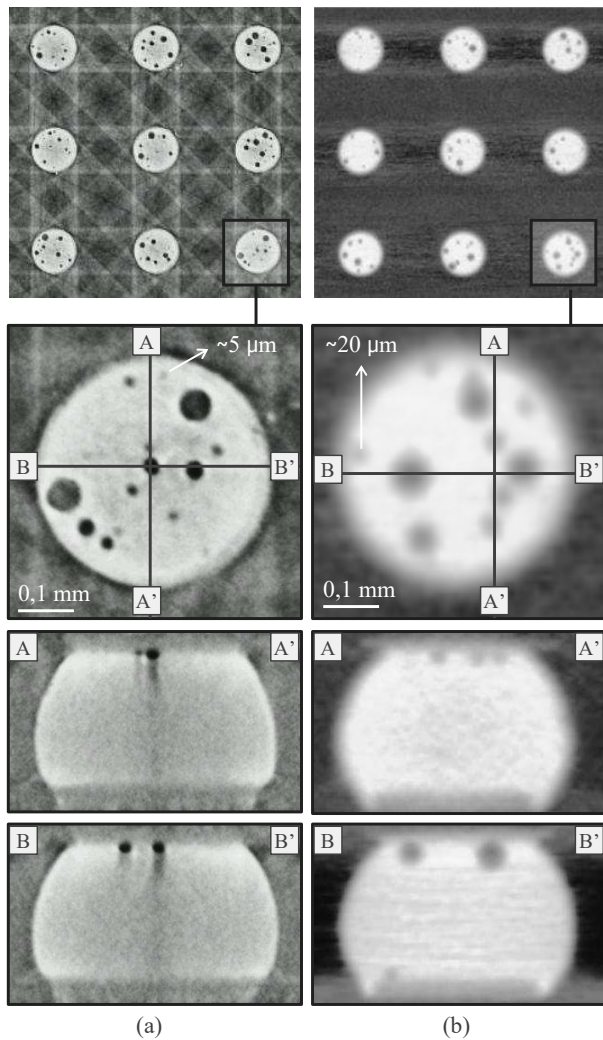


Fig. 7. Representative cross sections of a BGA-256 joint reconstructed using the synchrotron setup (a) and the UGCT laboratory system (b). In each subfigure, the horizontal section (parallel to the board) shows two perpendicular lines, AA' and BB' , which correspond to the vertical cross sections shown below.

strain amplification and the associated risk of crack initiation. Spatially resolved void information is therefore essential for physically meaningful fatigue simulations.

Third, although overall joint-to-joint volume variations are small, a clear location-dependent trend is observed for the larger BGA type: the central joints are slightly wider, lower in height, and overall smaller in volume than the peripheral ones. This systematic effect, consistently observed across samples, is attributed to PCB warpage during reflow and demonstrates that subtle process-induced geometric biases are present even in commercial assemblies. In addition, the extracted joint profiles do not consistently exhibit the classical barrel shape often assumed in reliability models. Since local stress and strain fields are sensitive to joint geometry, such idealized assumptions may introduce non-negligible modeling errors.

Finally, a positive correlation between joint volume and total void volume is observed. As the measured volume

differences mainly reflect variations in stencil-printed solder paste deposition—and thus in associated flux content—this trend is physically consistent with void formation driven by flux outgassing during reflow. Larger deposited paste volumes inherently increase the likelihood of gas entrapment prior to solidification. This coupling between geometric variability and void formation further supports the use of probabilistic approaches that account for correlated input parameters.

Taken together, these results highlight the necessity of incorporating realistic joint geometries, spatially varying void distributions, and their statistical variability into predictive models of solder joint performance in order to improve the reliability of fatigue life assessments for BGA assemblies.

ACKNOWLEDGMENT

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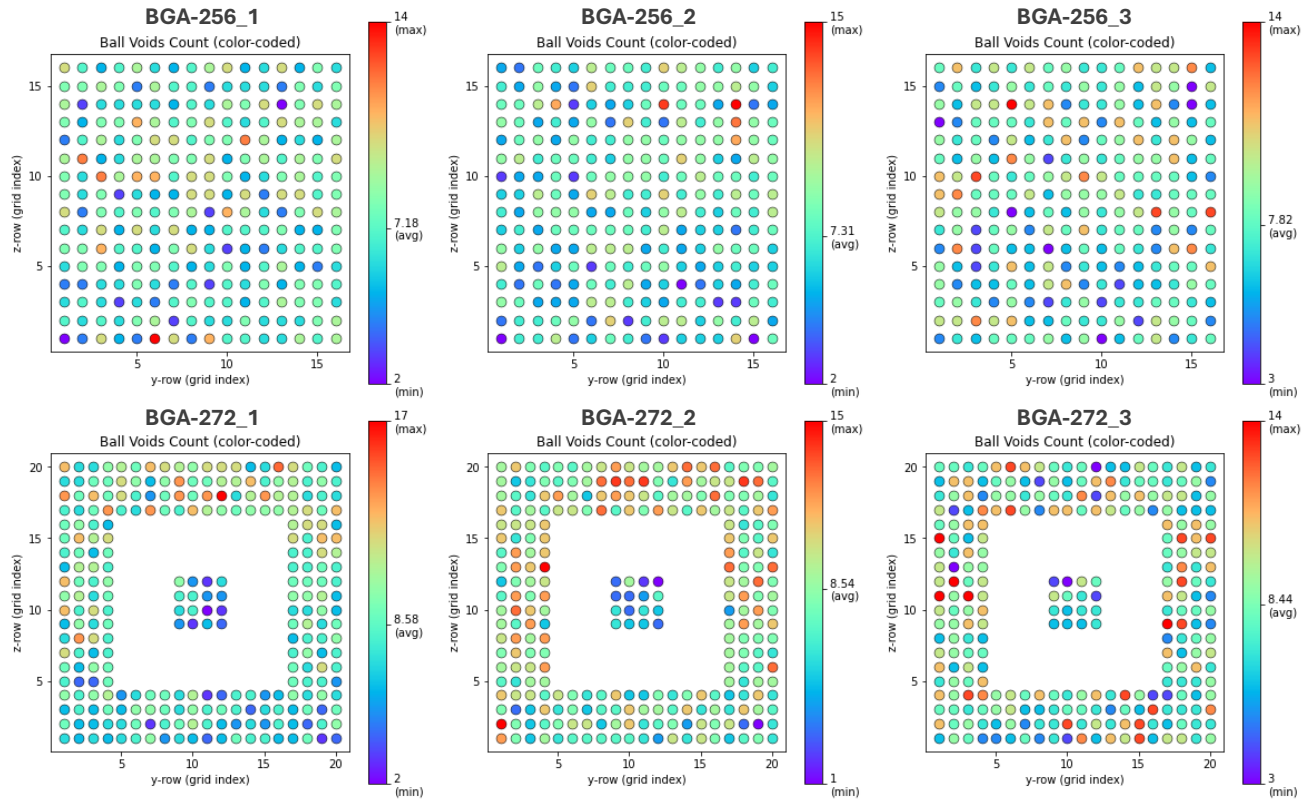


Fig. 8. Number of voids per BGA solder joint mapped onto their array positions. Each panel shows a different sample.

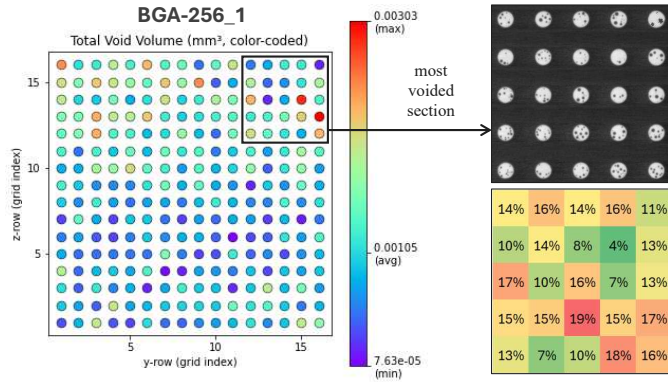


Fig. 9. Total void volume (mm^3) per BGA solder joint mapped onto their array positions for BGA-256_1. A 5×5 subset of balls is highlighted, for which the area fraction in the worst section of each ball was computed manually.

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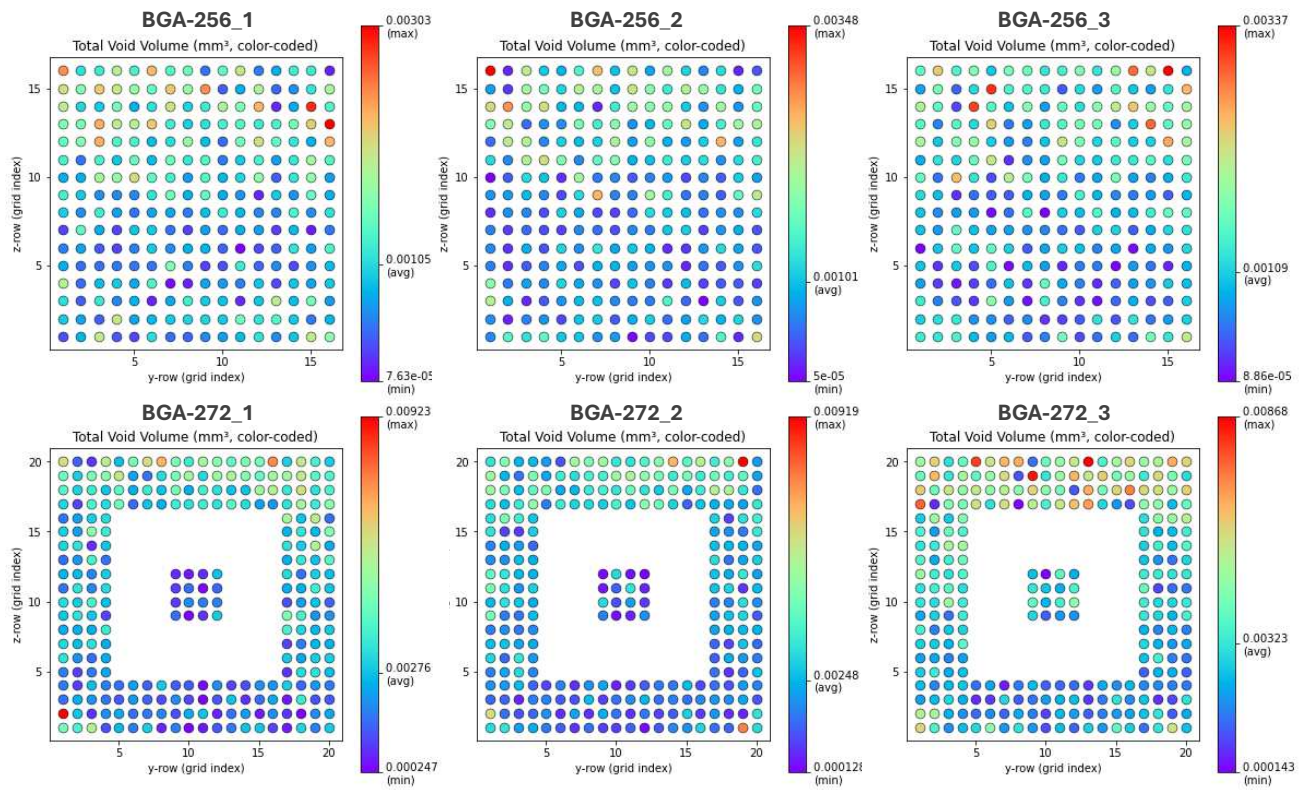


Fig. 10. Total void volume (mm^3) per BGA solder joint mapped onto their array positions. Each panel shows a different sample.

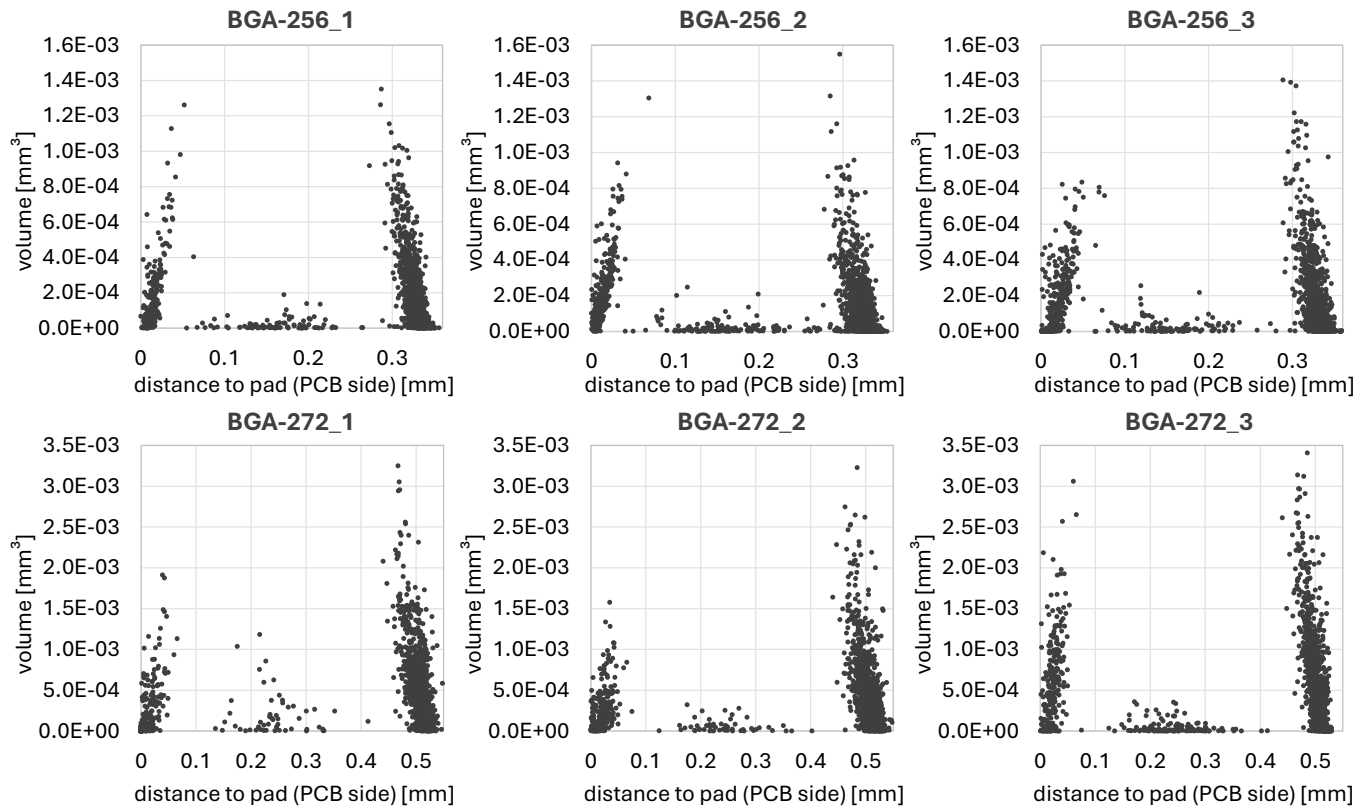


Fig. 11. Void volume versus distance from the PCB pad for each BGA solder joint. Each of the six panels represents a different sample.